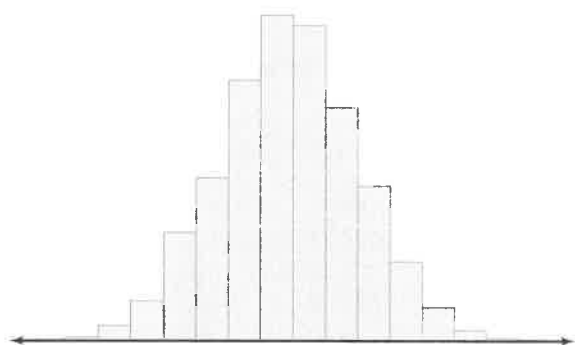


WORKSHEET

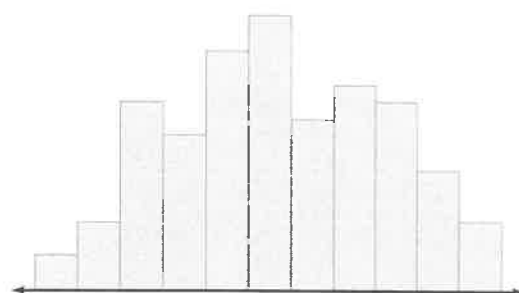
The normal curve

1 Copy or print each histogram and sketch a normal curve over each one.

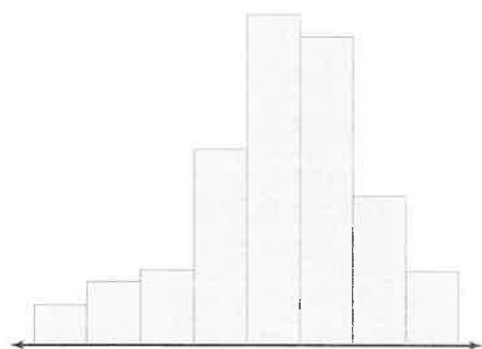
a



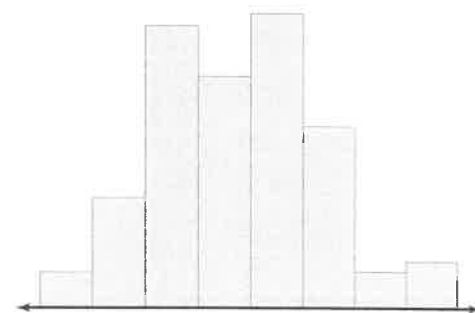
c



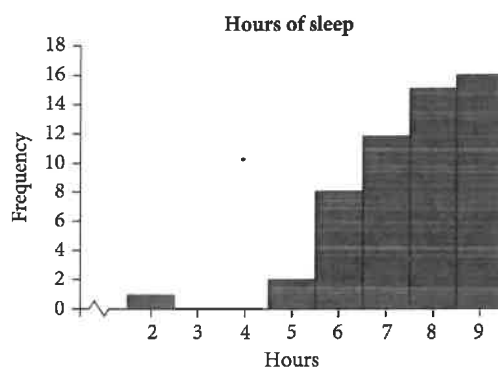
b



d



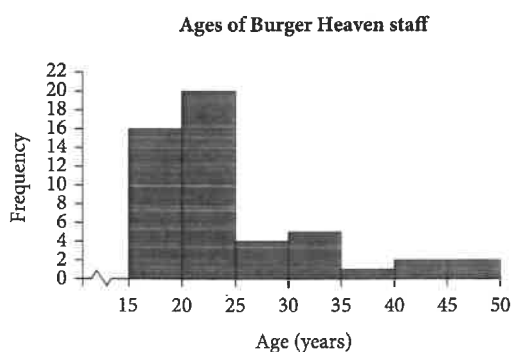
The following histogram relates to questions 2, 3 and 4.



- 2 The most appropriate description for the shape of the histogram is
 - A symmetrical
 - B bimodal
 - C positively skewed with an outlier
 - D negatively skewed with an outlier
 - E none of the above
- 3 The centre value for the number of hours sleep is
 - A 6
 - B 7
 - C 8
 - D 15
 - E 27
- 4 The spread of the data could be described as:
 - A Hours of sleep varies from 0 to 9 hours
 - B Frequency varies from 0 to 9 hours
 - C Hours of sleep varies from 5 to 9 hours
 - D Frequency varies from 5 to 9 hours
 - E Hours of sleep varies from 2 to 9 hours

Reasoning and communication

- 5 The following histogram displays the ages of 50 Burger Heaven staff members. Write a brief report describing the data displayed in this graph.



Exit Pass – Do all except #3 (we'll get to that later)

- 1 In your own words, describe what a back-to-back stem plot is. Draw a rough sketch of what one would look like.
- 2 What is a five number summary?
- 3 In your own words, describe what a double boxplot is. Draw a rough sketch of what one would look like.
- 4 Match the terms in the left column with their correct meanings in the right column.

bell-shaped	A distribution with one peak.
bimodal	A rectangular distribution where every score has a similar frequency.
normal	The symmetrical shape of a normal distribution.
skewed	A distribution with two peaks.
unimodal	The scores that divide the data into four equal parts.
quartiles	A distribution where most scores are to the left or right of centre and there is a tail on the side that doesn't have many scores.
uniform	A symmetrical distribution whose mean, median and mode are the same.